Notes: Clayton, Dos Santos, Maggiori, & Schreger (AER, 2025)

Internationalizing Like China

Contents

1 Big picture

- **Question.** How is China internationalizing its domestic bond market, why has it liberalized gradually rather than all at once, and what does this reveal about how a country builds credibility as a safe local-currency issuer?
- **Setting.** China's onshore bond market from the early 2000s to 2023, with institutional detail on the sequence of access programs — QFII, RQFII, CIBM Direct, and Bond Connect — and a model of crises, capital flight, and government reputation.
- **Core empirical facts.** Foreign participation in China's domestic bond market rose sharply, but the early rise came mainly from central banks and other stable official investors, while the later rise came from private investors. China deliberately engineered this pattern through quotas, lockups, and registration requirements that first favored stable investors and only later admitted flightier private capital.
- **Cross-border portfolio result.** In global mutual-fund and ETF portfolios, Chinese renminbi government bonds sit between developed-market and emerging-market bonds. They are not held purely like a standard EM asset, but neither are they yet held like core DM reserve assets.
- **Theory.** The paper models a government that may be tempted to impose ex post capital controls in crises, two classes of foreign investors (stable and flighty), and slow reputation building. Reputation is the perceived probability that the government will let foreigners repatriate capital in a crisis.
- **Main model result.** At low reputation, the government optimally borrows only from stable investors. Once reputation rises enough, it opens the market to flighty investors too. That opening is itself a costly signal, so it generates a jump in reputation.
- **Crisis application.** The 2015–2016 capital outflow episode is interpreted as a reputational test. Foreigners withdrew funds, but China did not lock them in. In the model, that choice generates a V-shaped pattern: capital leaves in the crisis, then returns more strongly once investors update upward on the government's type.
- **Main contribution.** The paper turns China's bond-market opening into a dynamic safe-asset and reputation problem. Internationalization is not treated as a static deregulation choice but as a gradual state-contingent process shaped by the investor base and by crisis behavior.
- **Why we care.** The paper is about more than China. It gives a framework for thinking about how any country that wants deeper foreign participation in local-currency debt must trade off cheaper external financing against the fragility created by capital flight.

2 Data and measurement (key objects)

2.1 Foreign ownership of China's domestic renminbi bond market

- **Main holdings series.** The paper combines official reserve data from IMF COFER with private portfolio data from IMF CPIS and commercial sources to measure foreign ownership of RMB-denominated bonds issued by China-resident entities.
- **Time pattern.** Foreign ownership of onshore RMB bonds rises from under \$150 billion in early 2014 to almost \$660 billion at the start of 2022, before falling back to about \$515 billion by late

2023 as China again experienced outflows.

- **Composition.** Early growth is dominated by reserve managers, especially central banks; only later, especially in 2019–2021, does private foreign investment become quantitatively important.
- **Asset composition within China.** Foreign investors mostly buy the safest onshore RMB instruments: China Government Bonds (CGBs) and policy bank bonds (PBBs), with very little exposure to the part of the market carrying meaningful private credit risk.

2.2 Institutional opening-up data and investor-type classification

- **Programs tracked.** The paper builds a monthly dataset of investor entry under QFII, RQFII, CIBM Direct, and Bond Connect.
- **Sources.** Entry timing comes from regulatory reports and access rosters. Investor identities are matched to FactSet for country, nationality, and industry classification.
- **Investor classes.** "Stable" investors include central banks, sovereign wealth funds, pension funds, insurance companies, international organizations, endowments, and nonprofits. "Flighty" investors are mostly portfolio-management firms such as mutual funds, ETFs, and hedge funds. "Banks" are tracked separately but play less of a conceptual role in the main model.
- **Why this matters.** The paper is not merely showing that foreign holdings rose; it documents that the Chinese government selected the *type* of foreign investor that could enter at each stage of liberalization.

2.3 Private portfolio holdings across countries and currencies

- **Micro data.** The paper uses security-level holdings of foreign mutual funds and ETFs domiciled in more than 50 countries, excluding China.
- **Sample restrictions.** The baseline analysis focuses on foreign-currency government bonds, excludes domestic-currency holdings, excludes currency specialists (more than 50% of foreign bond holdings in one currency), and excludes very small foreign-currency portfolios (less than $20 million).
- **Final sample.** The resulting sample contains 828 funds and just over $1.6 trillion in assets under management.
- **Key portfolio object.** The paper studies whether funds that hold RMB bonds otherwise look like developed-market bond funds or emerging-market bond funds.

2.4 Institutional background: what opening-up actually meant

- **Early stage.** QFII (from 2002) allowed limited, quota-based foreign access and had lengthy lockups.
- **Middle stage.** CIBM reforms in the 2010s opened direct access for official long-term investors such as central banks and sovereign wealth funds.
- **Late stage.** Bond Connect (from 2017) sharply reduced frictions for foreign private investors by routing access through Hong Kong and standard trading platforms.

- **Interpretation.** The reform path progressively lowered the cost of entry and repatriation. That is exactly what the model interprets as moving from a market designed for stable investors toward one also usable by flighty investors.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Portfolio-share statistics for RMB versus DM and EM bonds

For fund i and currency c, the share of the foreign-currency bond portfolio invested in currency c is

$$\alpha_{c,i} = \frac{\sum_{b \in B_c} MV_{b,i}}{\sum_{c \in FC_i} \sum_{b \in B_c} MV_{b,i}}. \tag{1}$$

The share of the *remaining* foreign-currency bond portfolio invested in developed-market currencies is

$$\alpha_{DM,c,i} = \frac{\sum_{d \in \{DM_i / c\}} \alpha_{d,i}}{1 - \alpha_{c,i}}. \tag{2}$$

The summary statistic of interest is the cross-sectional correlation across funds,

$$\rho_{c,DM} = \text{corr}_i\left[\alpha_{c,i}, \alpha_{DM,c,i}\right]. \tag{3}$$

- **Interpretation.** If a currency has a high positive $\rho_{c,DM}$, then funds that hold more of that currency also hold more developed-market government bonds. If $\rho_{c,DM}$ is strongly negative, that currency behaves like an emerging-market allocation.
- **What the paper finds.** RMB has an intermediate value: much more DM-like than typical EM currencies, but still not in the core group with the dollar, euro, yen, or pound.
- **Economic meaning.** The portfolio evidence is interpreted as a revealed-preference measure of China's current reputation in global fixed-income markets.

3.2 Stage-game environment: debt issuance, liquidation, and pledgeability

At the beginning of date t, the intermediary raises foreign debt and undertakes projects of scale

$$I_t = A + D_t, \qquad D_t = D_t^s + D_t^f. \tag{4}$$

In the middle of the period, if investors do not fully roll over debt, the intermediary liquidates long-term projects. Total rolled-over debt satisfies

$$D_t^\ell = R_t D_t - \gamma L_t, \tag{5}$$

and the rollover market is constrained by pledgeability,

$$R_t^\ell D_t^\ell \leq (1 - h_t)(Q I_t - L_t). \tag{6}$$

End-of-date intermediary equity payoff is

$$c_t = QI_t - L_t - R_t^\ell D_t^\ell, \tag{7}$$

which can be rewritten as

$$c_t = \frac{h_t}{\gamma - \dfrac{1-h_t}{R_t^\ell}} \left(\gamma Q I_t - R_t D_t\right). \tag{8}$$

- **Interpretation.** Capital flight forces liquidation. The worse are liquidation discounts and pledgeability conditions, the larger is the destruction of intermediary equity.
- **Why this matters.** Ex post capital controls are tempting because they keep funds from leaving and therefore reduce forced liquidations.

3.3 Stable versus flighty investors

Pledgeability depends on whether the country has admitted flighty investors:

$$h_t = \begin{cases} h^s, & D_t^f = 0, \\ h^f, & D_t^f > 0, \end{cases} \qquad h^f \geq h^s. \tag{9}$$

- **Interpretation.** Flighty investors require the intermediary to maintain a lower debt-to-asset ratio in order to roll over debt.
- **Economic idea.** Opening the market to private portfolio investors makes crisis deleveraging more acute for the *whole* market, not just for the marginal entrant.
- **Institutional mapping.** This pari passu assumption is meant to capture the fact that reforms like Bond Connect changed the market environment broadly rather than creating a narrow side window used only by one investor class.

3.4 Investor problem and the interest-rate schedules

At the beginning of date t, an investor of type $i \in \{s, f\}$ chooses lending D_t^i to solve

$$\max_{D_t^i \geq 0} \Big\{ \bar{R}w + (R_t^i \mathbb{E}[1-\tau] - \bar{R}) D_t^i - \frac{1}{4} \frac{b}{\omega(M_t)} (D_t^i)^2 \Big\}. \tag{10}$$

The equilibrium of the rollover market and the investor's beginning-of-period optimization imply

$$R_t^i = \frac{\bar{R} + \dfrac{1}{2} \dfrac{b}{\omega(M_t)} D_t^i}{1 - (1 - M_t)\bar{\tau}}, \tag{11}$$

$$R_t^\ell = 1 - \tau. \tag{12}$$

- **Interpretation.** M_t is the government's reputation for not imposing ex post capital controls. A higher M_t lowers the intercept and slope of the interest-rate schedule faced by the issuer.

- **Core mechanism.** The better the reputation, the cheaper and more elastic is foreign funding ex ante. That is why governments want to build reputation.

3.5 Committed government's optimal issuance policy and the opening-up threshold

The committed government chooses debt issuance to maximize its static payoff, taking investors' beliefs as given. Proposition 1 yields

$$D^s(M_t) = \frac{\omega(M_t)}{b}\Big(\gamma Q\left[1-(1-M_t)\bar{\tau}\right] - \bar{R}\Big), \tag{13}$$

$$D^f(M_t) = \begin{cases} 0, & M_t \leq M^*, \\ D^s(M_t), & M_t > M^*, \end{cases} \tag{14}$$

and the resulting equilibrium interest rate is

$$R(M_t) = \frac{1}{2}\frac{\bar{R}}{1-(1-M_t)\bar{\tau}} + \frac{1}{2}\gamma Q, \qquad R(M_t) = R^s(M_t) = R^f(M_t). \tag{15}$$

The indirect utility of the committed government is

$$V(M_t) = \frac{h(M_t)}{\gamma - [1-h(M_t)]}\Big[\gamma Q I(M_t) - R(M_t)D(M_t)\Big]. \tag{16}$$

- **Interpretation.** There is a unique threshold M^* at which the government starts borrowing from flighty investors as well as stable ones.
- **Why a threshold appears.** Letting in flighty investors expands the pool of capital, but it also worsens the haircut from h^s to h^f and therefore raises the cost of crises. The government opens up only once the reputational gains are large enough to justify that fixed cost.
- **Key intuition.** Low-reputation governments want stable investors because cheap funding is not yet large enough to compensate for the fragility created by flighty investors.

3.6 Opportunistic government, capital controls, and reputation

An opportunistic government has payoff

$$V^{\mathrm{Opp}}(M_t, \tau) = \begin{cases} V(M_t), & \tau = 0, \\ g(M_t)V(M_t), & \tau = \bar{\tau}, \end{cases} \tag{17}$$

where

$$g(M_t) = \frac{\gamma - [1-h(M_t)]}{\gamma - \dfrac{1-h(M_t)}{1-\bar{\tau}}}. \tag{18}$$

Investor beliefs about the government's type generate the reputation mapping

$$M(\pi_t) = \pi_t + (1 - \pi_t)m(\pi_t), \tag{19}$$

where $m(\pi_t)$ is the probability investors assign to an opportunistic government not imposing capital controls.

- **Interpretation.** The opportunistic type likes controls because they limit liquidation losses. But using controls reveals the type and damages future borrowing terms.
- **Why $g(M_t)$ falls with flighty investors.** Once the market has opened to flighty investors, the marginal benefit of imposing controls is lower relative to the reputational cost, because the market structure has changed and the government has more to lose from being revealed as opportunistic.

3.7 Dynamic reputation building

If the government does *not* impose controls in period t, Bayes' rule implies

$$\pi_{t+1} = \epsilon^O + (1 - \epsilon^C - \epsilon^O)\frac{\pi_t}{M(\pi_t)}. \tag{20}$$

If it *does* impose controls, then $\pi_{t+1} = \epsilon^O$.

The opportunistic government's Bellman problem is

$$\begin{aligned} W(\pi_n) = \max_{m_n^o \in [0,1]} \; & m_n^o \Big[V^{\mathrm{Opp}}(M(\pi_n), 0) + \beta W(\pi_{n+1})\Big] \\ & + (1 - m_n^o)\Big[V^{\mathrm{Opp}}(M(\pi_n), \bar{\tau}) + \beta W(\pi_0)\Big]. \end{aligned} \tag{21}$$

For a mixed strategy to be optimal, the path of indirect utility must satisfy

$$V(M_{n+1}) = \frac{g(M_n)}{g(M_{n+1})}\rho(M_n)V(M_n) + \frac{g(M_0)}{g(M_{n+1})}V(M_0), \tag{22}$$

with

$$\rho(M_n) = \frac{1}{\beta}\frac{g(M_n) - 1}{g(M_n)}. \tag{23}$$

- **Interpretation.** Reputation evolves along an indifference path: the opportunistic government must be just willing to mimic the committed type and refrain from imposing controls today in order to keep borrowing advantages tomorrow.
- **Graduation step.** The cycle ends at a "graduation" step N where opportunistic governments stop mimicking and impose controls for sure.
- **Opening-up step.** The earliest step at which the government admits flighty investors is N^*. In the model, opening up and graduation are separate objects.

3.8 Homogeneous-investor benchmark

If all investors are equally flighty, $h^s = h^f$, the indifference path simplifies to

$$V(M_{n+1}) = \rho^f V(M_n) + V(M_0), \tag{24}$$

with graduation when

$$V(1 - \epsilon^C) \leq \rho^f V(M_N) + V(M_0). \tag{25}$$

- **Interpretation.** Without heterogeneity, reputation builds smoothly and there is a unique graduation-step Markov equilibrium.
- **Usefulness.** This case gives the basic logic before the paper reintroduces the richer stable-versus-flighty investor structure.

3.9 Heterogeneous investors and the opening-up jump

Before the market opens to flighty investors, the path is

$$V(M_{n+1}) = \rho^s V(M_n) + V(M_0), \qquad n + 1 < N^*. \tag{26}$$

At the opening-up step,

$$V(M_{N^*}) = \frac{g^s}{g^f} \Big[\rho^s V(M_{N^*-1}) + V(M_0) \Big]. \tag{27}$$

After opening up but before graduation,

$$V(M_{n+1}) = \rho^f V(M_n) + \frac{g^s}{g^f} V(M_0), \qquad N^* \leq n < N. \tag{28}$$

- **Interpretation.** Opening up is a costly signal. It causes a discrete jump in reputation because it is disproportionately expensive for an opportunistic government to mimic.
- **Economic implication.** Capital inflows jump up at opening for two reasons: the market suddenly admits a second investor class, and the issuer's reputation itself jumps.
- **Multiplicity point.** The paper notes that, with heterogeneous investors, there can be multiple equilibria with different opening-up steps, although for any fixed opening-up step there is at most one graduation-step Markov equilibrium.

3.10 High-state / low-state extension and the crisis test

The crisis extension adds a high state with probability p and a low state with probability $1 - p$. In the extended setup,

$$\mathfrak{M}_t = p + (1 - p) M_t, \tag{29}$$

and the beginning-of-period interest rate becomes

$$R_t = \frac{\bar{R} + bD_t^i/2}{1 - (1 - \mathfrak{M}_t)\bar{\tau}}. \tag{30}$$

Proposition 4 states that, for a government inside the reputation cycle,

$$\text{high state: } D_t = D_{t+1}, \qquad D_t^\ell = R_t D_t, \tag{31}$$

$$\text{low state and } \tau = 0: \quad D_t < D_{t+1}, \qquad D_t^\ell\big|_{\tau=0} < R_t D_t, \tag{32}$$

$$\text{low state and } \tau = \bar{\tau}: \quad D_t > D_{t+1}, \qquad D_t^\ell\big|_{\tau=\bar{\tau}} > D_t^\ell\big|_{\tau=0}. \tag{33}$$

- **Interpretation.** Good times do not reveal much about type. Crises do. If the government lets foreigners leave, reputation rises and capital returns stronger later. If the government traps capital, it enjoys a short-run gain but suffers a long-run reputational collapse.
- **Empirical mapping.** This is the formal explanation for the V-shaped capital-flow response around the 2015–2016 China episode.

4 Identification and credibility

4.1 What is identified empirically

- **Not an IV paper.** The empirical part is primarily a characterization paper rather than a reduced-form causal design with an instrument or natural experiment.
- **Core empirical leverage.** The evidence comes from policy sequencing, investor-entry timing, asset-holding composition, and cross-country portfolio correlations.
- **Why this is still informative.** The institutional design of the opening-up programs was explicit about quotas, lockups, and eligible investor categories, so the investor-base result is not a loose interpretation of raw inflows; it is tied directly to policy rules.

4.2 Why the "selection of the investor base" claim is credible

- **Policy design.** QFII and related schemes were quota-based and restrictive, while later reforms like Bond Connect greatly lowered access frictions. This gives the paper a clean institutional narrative for why stable investors entered first and flightier private investors entered later.
- **Data construction.** The paper does not infer investor type from noisy flow behavior alone. It constructs monthly entry lists and classifies institutions directly using matched investor identities.
- **Visual evidence.** Figure 2 shows stable investors entering much earlier in the cumulative distribution than flighty investors, with flighty entry accelerating only after the later liberalization wave and bond-index inclusion.

4.3 Why the portfolio interpretation is credible

- **Security-level holdings.** The currency-composition exercise is based on actual mutual-fund and ETF holdings, not country-level aggregates.
- **Clean sample.** The baseline keeps only government bonds, excludes domestic currency holdings, and drops specialist funds, which makes the RMB-versus-DM/EM comparison much easier to interpret.
- **Meaning of the result.** The paper is careful not to claim that the correlation statistic is a sufficient statistic for safe-asset status. Instead, it treats it as disciplined suggestive evidence that global investors see China as an intermediate case.

4.4 Why the model is believable

- **Direct mapping to the institutions.** Stable investors map to central banks, sovereign wealth funds, and other long-horizon institutions; flighty investors map to mutual funds, ETFs, and hedge funds.
- **Direct mapping to investor fears.** The key reputational object is repatriation risk — whether foreign investors can "get their money out" during a crisis — which is exactly the concern highlighted by practitioners during the 2015–2016 episode.
- **Why a reputation model is natural.** The policy question is about an issuer with potentially large long-run ambitions but uncertain crisis behavior. That is exactly the environment where repeated-game reputation logic is useful.

4.5 Limitations

- **No quasi-experimental causal claim.** The paper does not identify the causal effect of a given reform on a specific outcome with an external instrument.
- **Model is stylized.** The theory abstracts from many other crisis policies, such as reserve accumulation, bank bailouts, outright default, or exchange-rate depreciation, even though the authors discuss these extensions.
- **Multiplicities are possible.** With heterogeneous investors, the theory can feature multiple equilibria associated with different opening-up steps.
- **Still persuasive overall.** The descriptive facts line up tightly with the model's mechanism, and the theory is designed to rationalize exactly the observed sequencing, portfolio patterns, and crisis behavior.

5 Tables and figures

5.1 Figure 1: Composition of foreign ownership of China-issued RMB bonds

Read:

- The stacked bars show that foreign ownership grows strongly from 2014 to 2021, but the early increase is dominated by central bank reserves rather than private investors.

- Reserve holdings remain the core of the stock through 2018–2020, while private investment becomes quantitatively important only in the later years.
- The 2021 decomposition shows that private holdings are geographically broad-based, with notable participation from Hong Kong, the euro area, the United States, Singapore, and Great Britain.
- This is the first key fact of the paper: China's bond-market internationalization started with stable official inflows, not with hot private money.

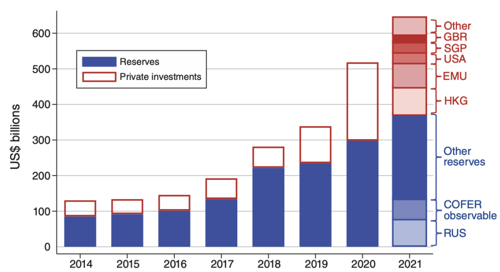

Figure 1. Composition of Foreign Ownership of China-Issued RMB Bonds

Notes: The figure plots our estimated breakdown of foreign ownership of RMB-denominated bonds into central bank reserves and private holdings. Data on reserves are from IMF COFER, and private holdings are from IMF CPIS or from commercial data. See Supplemental Appendix A.I.A for details.

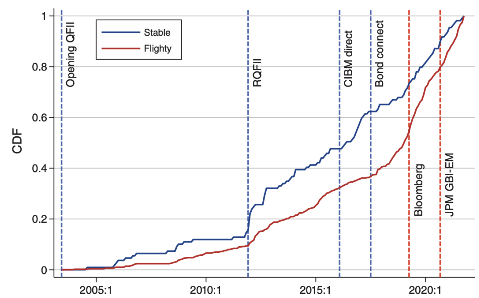

Figure 2. Entry into Domestic Markets

Notes: The figure plots the share of each investor type that had entered the market by a given date. The share is expressed as a fraction of investors by type that had entered by 2021.

5.2 Figure 2: Entry into domestic markets

Read:

- The blue line (stable investors) lies above the red line (flighty investors) for most of the sample, meaning stable investors entered earlier and accumulated access faster.
- There are visible jumps in stable participation around the RQFII and CIBM Direct reforms.
- Flighty participation accelerates later, especially after Bond Connect and major bond-index inclusion milestones.
- The figure is the clearest exhibit that China consciously staggered the entry of investor types.

5.3 Figure 3: Portfolio shares by currency

Read:

- Panel A (Brazilian real) shows a negative relation between BRL holdings and DM holdings: BRL debt is held mainly by EM-focused funds.
- Panel C (Japanese yen) shows the opposite pattern: JPY debt is held mainly by DM-focused funds.
- Panel B (RMB) is roughly flat and in the middle. RMB government bonds are held by both kinds of funds.
- The highlighted T. Rowe Price and PIMCO funds make the intuition vivid: they look very different in BRL and JPY, but much more similar in RMB.

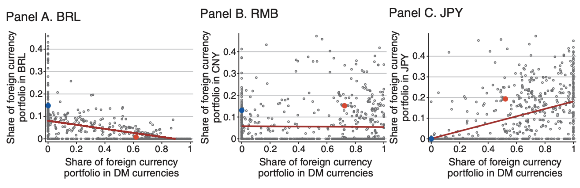

FIGURE 3. PORTFOLIO SHARES BY CURRENCY

Notes: In each panel, observations are the portfolio holding shares of a particular fund in December 2020. The vertical axis is the portfolio share in Brazilian reais in the left panel, RMB in the middle panel, and Japanese yen in the right panel. The horizontal axis in all panels is the portfolio share in developed markets currencies. In each panel, the blue dot represents the holdings of the PIMCO Emerging Markets Local Currency and Bond Fund, and the red dot represents the holdings of the T. Rowe Price International Bond Fund. Data from December 2020.

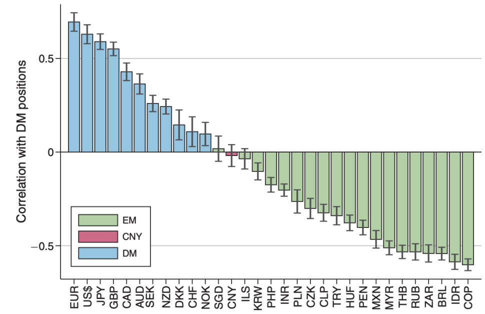

FIGURE 4. PORTFOLIO SIMILARITY WITH DEVELOPED COUNTRIES' LOCAL CURRENCY GOVERNMENT BONDS

Notes: The figure reports the correlation between the holdings of bonds in each currency and holdings in developed markets (DM) currencies. The set of funds for measuring the correlation are restricted to nonspecialists (less than 50 percent of its AUM in any single foreign currency) and those that have more than $20 million of foreign currency investment. Gray lines correspond to 95 percent confidence intervals computed via bootstrapping.

5.4 Figure 4: Portfolio similarity with developed countries' local-currency government bonds

Read:

- Core DM currencies like the euro, dollar, yen, and pound have high positive correlations with DM shares. Standard EM currencies have negative correlations.
- RMB sits between those groups and close to the most developed of the EM currencies, near Singapore, Israel, and South Korea.
- This figure summarizes the paper's revealed-preference portfolio argument in one chart: China is no longer a standard EM bond market, but it is not yet a full reserve-asset market either.

5.5 Figure 5: Model timeline within each date

Read:

- The figure lays out the stage game: at the beginning of the date the country chooses project scale and foreign debt; in the middle it chooses capital controls, liquidation, and rollover; at the end payoffs are realized.
- The timeline shows exactly where the tension lives: ex ante the government wants cheap foreign funding, but ex post it may want to trap foreign capital when crises hit.
- This timing is essential for the reputation mechanism because the same government that borrows before the crisis decides whether to impose controls during it.

5.6 Figure 6: Equilibrium reputation cycle with heterogeneous foreign investors

Read:

- Panel A shows reputation M_n and beliefs π_n rising over time, with a discrete upward jump at the opening-up step N^*.
- Panel B shows the mimicking probability of the opportunistic government declining to zero at the graduation step N.

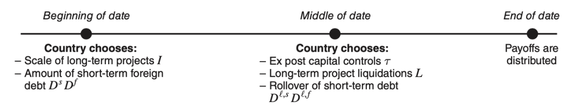

FIGURE 5. MODEL TIMELINE WITHIN EACH DATE

Note: The figure displays the timing of the stage game within each date *t*.

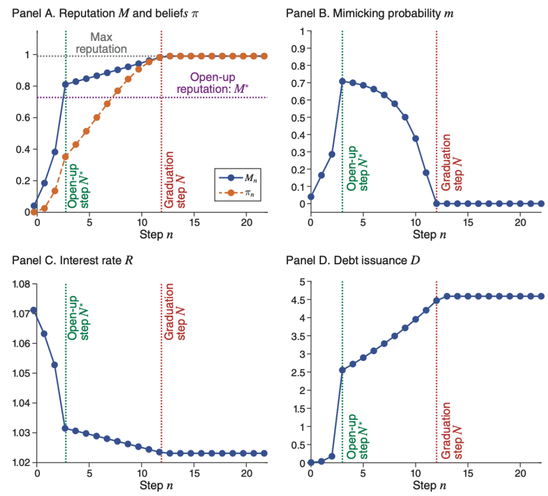

FIGURE 6. EQUILIBRIUM REPUTATION CYCLE: HETEROGENEOUS FOREIGN INVESTORS

Notes: Numerical illustration of the equilibrium of the model when foreign investors are heterogeneous. The N^* dashed green and N dashed red lines are the opening-up and graduation steps, respectively.

- Panel C shows the interest rate falling as reputation rises. Panel D shows debt issuance jumping at opening up and then continuing to grow until graduation.
- The figure is the model's main visual punchline: gradual opening and gradual reputation building go together, and opening itself is a signaling event.

5.7 Figure 7: V-shaped recoveries and reputation crashes

Read:

- The green path is the no-crisis benchmark: debt stays flat because nothing is learned.
- The blue path is the good crisis response: foreigners flee in the middle of the date, but the government does not impose controls; reputation rises; debt later rebounds above the initial level.
- The red path is the bad crisis response: controls reduce the short-run drop in debt but destroy reputation, pushing future debt down to the low-reputation level D_0.
- This figure distills the whole logic of the paper. Capital controls are attractive in the short run but destructive for internationalization in the long run.

5.8 Figure 8: Foreign ownership of China's domestic bonds in the 2015–2016 capital flight

Read:

- The figure plots foreign holdings in RMB and in U.S. dollars. Both series fall sharply into early 2016 and then rebound quickly.
- In RMB terms, foreigners sell about 11% of their holdings between 2015:II and 2016:I; in dollar terms the decline is about 15% because valuation effects add to the drop.

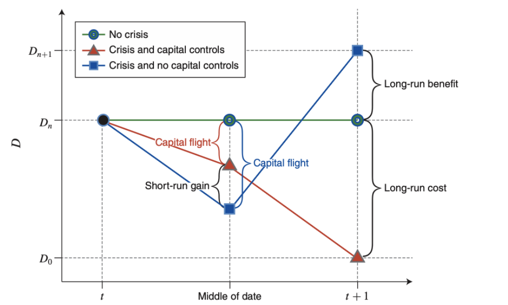

FIGURE 7. V-SHAPED RECOVERIES AND REPUTATION CRASHES

Notes: Illustration of the foreign debt dynamics of the model over a date t and reputation step n. The figure is purely illustrative and not drawn to scale in the interest of easier visualization.

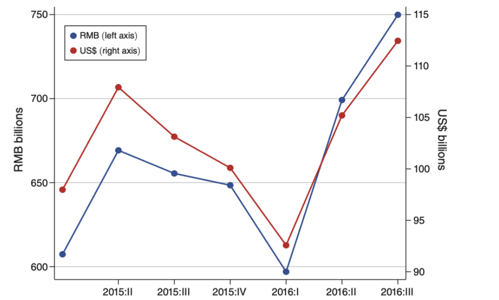

FIGURE 8. FOREIGN OWNERSHIP OF CHINA'S DOMESTIC BONDS IN THE 2015–2016 CAPITAL FLIGHT

Notes: This figure reports foreign holdings of China's domestic bonds valued in RMB (left axis) and US dollars (right axis). Data are from Shanghai Clearing House (SHCH) and China Central Depository and Clearing (CCDC).

- By 2016:II, holdings are above the prior high and continue rising. That is the empirical V-shape the model is designed to interpret.
- The key reading is not just "China had outflows." It is "China had outflows, did not trap foreigners, and then got stronger inflows later."

6 Short summary

- **Conclusion.** China internationalized its bond market by first admitting stable foreign investors and only later admitting flightier private investors.
- **Intuition.** A government that wants to become a safe-asset issuer must convince foreigners that it will not impose ex post capital controls in crises. Stable investors are useful early because they provide foreign participation without creating as much crisis fragility.
- **Empirical lesson.** The investor base matters as much as the size of inflows. Official inflows, private portfolio allocations, and crisis behavior all reveal where a country sits in the slow process of building financial reputation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** China's opening of its domestic bond market was deliberately sequential: stable investors first, flightier investors later.
- **Main theoretical message.** The sequencing is rational if the country is trying to build a reputation as a safe issuer while limiting the damage from capital flight.
- **Main empirical interpretation.** Global portfolio allocations suggest that China currently occupies an intermediate status between developed-market safe debt and standard emerging-market debt.
- **Crisis interpretation.** The 2015–2016 outflow episode is best understood as a reputational stress test that China passed by not trapping foreign capital.

- **Broader implication.** Internationalizing a bond market is not just about removing barriers; it is about choosing an investor base and then behaving consistently in crises so that the investor base is willing to deepen.

7.2 Economic intuition

- A low-reputation issuer benefits from foreign capital, but it also fears that foreigners will run in bad times.
- Stable investors are attractive at the beginning because they help build market participation without generating as much rollover risk.
- Once a government has demonstrated enough restraint in crises, it can safely broaden access to flightier investors, and doing so can itself strengthen beliefs that the market is becoming safer.

7.3 Takeaway

China's bond-market opening is best understood as a dynamic reputation problem. The paper's core insight is that becoming an international safe-asset provider is not a one-shot deregulation choice but a long sequence of policy commitments tested in crises. China has advanced meaningfully along that path, but the paper is careful that the end point is still uncertain: the real question is not whether foreign money has entered, but whether China will continue to leave when times are bad without shutting the gate.